quantum mystery, and leaves us dazzled by the paradoxical nature of reality that it exposes.

Although quantum mechanics had made the textbooks by the early 1930s, it is typical of Feynman that, as a young man, he preferred to refashion the theory for himself in an entirely new guise. The Feynman method has the virtue that it provides us with a vivid picture of nature's quantum trickery at work. The idea is that the path of a particle through space is not generally well defined in quantum mechanics. We can imagine a freely moving electron, say, not merely traveling in a straight line between A and B as common sense would suggest, but taking a variety of wiggly routes. Feynman invites us to imagine that somehow the electron explores all possible routes, and in the absence of an observation about which path is taken we must suppose that all these alternative paths somehow contribute to the reality. So when an electron arrives at a point in space—say a target screen—many different histories must be integrated together to create this one event.

Feynman's so-called path-integral, or sum-over-histories approach to quantum mechanics, set this remarkable concept out as a mathematical procedure. It remained more or less a curiosity for many years, but as physicists pushed quantum mechanics to its limits—applying it to gravitation and even cosmology—so the Feynman approach turned out to offer the best calculational tool for describing a quantum universe. History may well judge that, among his many outstanding contributions to physics, the path-integral formulation of quantum mechanics is the most significant.

Many of the ideas discussed in this volume are deeply philosophical. Yet Feynman had an abiding suspicion of philosophers. I once had occasion to tackle him about the nature of mathematics and the laws of physics, and whether abstract mathematical laws could be considered to enjoy an independent Platonic existence. He gave a spirited and skillful description of why this indeed appears so but soon backed off when I pressed him to take a specific philosophical position. He was similarly wary when I attempted to draw him out on the subject of reductionism. With hindsight, I believe that Feynman was not, after all, contemptuous of philosophical problems. But, just as he was able to do fine mathematical physics without systematic mathematics, so he produced some fine philosophical insights without systematic philosophy. It was formalism he disliked, not content.

It is unlikely that the world will see another Richard Feynman. He was very much a man of his time. The Feynman style worked well for a subject that was in the process of consolidating a revolution and embarking on the far-reaching exploration of its consequences. Postwar physics was secure in its foundations, mature in its theoretical structures, yet wide open for kibitzing